BUSINESS REQUIREMENTS DOCUMENT

Enterprise Banking Intelligence & Early Warning Platform (EBIEWP)

Client: Vertex Bank Plc (Fictional)
Document Status: Final
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1. Background

Vertex Bank Plc serves approximately 100,000 customers across Nigeria through its retail, SME, corporate and private banking divisions.

Management requires a centralized Business Intelligence capability that can transform operational banking data into trusted, decision-ready insight. The platform is designed to support proactive monitoring of customer behaviour, lending performance, branch performance and customer service activity, while identifying early warning signals that may require intervention.

The completed EBIEWP solution combines enterprise data warehousing, analytical modelling, Power BI reporting, Customer 360 intelligence, rules-based Early Warning Intelligence and an operational automation layer for stakeholder notification.

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2. Business Problem

The bank lacks a centralized intelligence capability that connects customer, financial, credit, branch and service information into a unified view and proactively identifies customers and business conditions requiring attention.

This can result in:

- Customer attrition and declining customer engagement
- Dormant or deteriorating customer relationships
- Increasing loan delinquency and repayment deterioration
- Limited visibility into customer-level financial and service risk
- Branch performance issues being identified reactively
- Increasing customer complaints and service deterioration
- Limited visibility into the reasons behind identified risks
- Delayed routing of identified issues to responsible stakeholders

- Reliance on manual reporting and fragmented operational analysis

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3. Proposed Solution

Develop and implement the Enterprise Banking Intelligence & Early Warning Platform (EBIEWP) as an end-to-end Business Intelligence and decision-intelligence solution.

The platform consolidates customer, account, transaction, loan, repayment and complaint data into an enterprise SQL Server data warehouse. The data is transformed into analytical structures and consumed through a Power BI semantic model and executive reporting layer.

Customer 360 and rules-based Early Warning Intelligence identify and prioritize customers requiring attention based on financial, credit and service indicators. The solution assigns warning reasons, severity, warning types, recommended actions and responsible stakeholders.

The completed solution extends the analytical workflow into operational automation using n8n, enabling stakeholder-specific early warning notifications through email.

Overall business flow:

Detect → Understand → Prioritize → Recommend → Route → Notify → Act

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4. Stakeholder Analysis

Stakeholder	Responsibilities	How They Will Use the Platform
Chief Executive Officer (CEO)	Oversees overall bank performance	Monitor overall bank health, executive KPIs, financial exposure and significant early warning indicators.
Head of Retail Banking	Responsible for retail customer growth and retention	Monitor customer behaviour, customer health and customers requiring proactive engagement.
Head of Credit Risk	Oversees the bank’s loan portfolio	Monitor repayment deterioration, Days Past Due, missed payments, loan exposure and credit-related warnings.
Regional Managers	Supervise multiple branches	Compare branch performance across regions

Stakeholder	Responsibilities	How They Will Use the Platform
		and identify branches or customer trends requiring management attention.
Branch Managers	Manage daily branch operations	Monitor branch KPIs, customer activity, complaints and relationship-management alerts associated with their branch.
Relationship Managers	Manage customer relationships	Receive and act on alerts for customers requiring proactive relationship engagement.
Customer Service	Manage customer complaints and service experience	Receive service-related alerts and follow up on customers showing complaint or service deterioration.
Service Recovery Team	Manage escalated service-recovery cases	Receive service-recovery alerts requiring targeted intervention.

5. Business Objectives

1. Provide a centralized view of banking performance across customer, account, lending, repayment, complaint and branch domains.
2. Provide management with trusted, business-ready data for decision-making.
3. Identify customers showing early signs of financial, credit or service deterioration.
4. Monitor loan and repayment behaviour and identify customers requiring proactive credit intervention.
5. Evaluate branch performance using operational and financial indicators.
6. Provide Customer 360 intelligence that combines financial, credit and service information.
7. Explain identified warnings through warning reasons, warning types and severity classifications.
8. Recommend appropriate business actions for identified warning conditions.
9. Route alerts to the appropriate operational stakeholder.
10. Demonstrate automated stakeholder notification as the final operational layer.

6. Business Value

EBIEWP is intended to improve the bank's ability to move from reactive reporting toward proactive, data-driven intervention.

The platform supports value creation by improving visibility into customer relationships, identifying potential credit and service deterioration earlier, highlighting operational performance issues, prioritizing intervention based on business rules and financial exposure, and reducing the delay between analytical detection and stakeholder action.

The solution uses rules-based Early Warning Intelligence rather than machine-learning prediction models. Its value therefore comes from transparent, explainable business logic that can be understood and acted upon by business users.

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7. Key Performance Indicators (KPIs)

7.1 Customer KPIs

- Total Customers
- Active Customers
- Dormant Customers
- Customers At Risk
- Customer Health
- Customer Attrition / Deterioration Indicators

7.2 Loan & Credit KPIs

- Total Loans
- Outstanding Loan Value / Principal
- Delinquent Loans
- Days Past Due (DPD)
- Consecutive Missed Payments
- Payment Behaviour
- Credit Warning Indicators

7.3 Branch KPIs

- Branch Customer Base
- Branch Financial Performance
- Branch Activity
- Branch Complaint Volume
- Branch Performance Indicators

7.4 Customer Experience KPIs

- Total Complaints
- Unsatisfied Complaints
- Repeat Complaints
- Complaint Resolution / Service Indicators
- Service Deterioration Indicators

7.5 Executive & Early Warning KPIs

- Overall Banking Health Indicators
- High-Severity Alerts
- Medium-Severity Alerts
- At-Risk Customers
- Watchlist Customers
- Total Outstanding Principal Associated with Alerts
- Recommended Interventions

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8. Project Scope

8.1 In Scope

- Customer analytics and segmentation
- Customer 360 intelligence
- Customer health monitoring
- Customer attrition and dormancy indicators
- Loan portfolio monitoring
- Repayment behaviour analysis
- Loan default-risk indicators based on predefined business rules
- Branch performance monitoring
- Customer complaint analytics
- Executive Business Intelligence dashboards
- Early Warning Intelligence
- Warning reason and severity classification
- Recommended business actions
- Stakeholder routing
- Automated stakeholder email notification using n8n

8.2 Out of Scope

- Fraud detection
- ATM monitoring

- Treasury operations
- Human Resources analytics
- Foreign exchange analytics
- Live integration with banking core systems
- Live transaction processing
- Machine-learning prediction models
- Production case-management integration
- Production-grade persistent alert deduplication and intervention tracking

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9. Business Questions

9.1 Customer Intelligence

- Which customers show early signs of deterioration?
- Which customers are At Risk or on the Watchlist?
- Which customers require immediate relationship engagement?
- What financial, credit or service signals explain the warning?
- Which customer segments demonstrate different behavioural patterns?

9.2 Loan & Credit Intelligence

- Which customers have overdue repayments?
- Which customers show consecutive missed payments?
- Which customers have elevated Days Past Due?
- Which customers have significant outstanding principal exposure?
- Which customers require proactive credit intervention?

9.3 Branch & Operational Intelligence

- Which branches are underperforming?
- Which branches have customer or operational deterioration?
- Which branches generate high complaint volumes?
- Which branches have customers requiring intervention?

9.4 Customer Experience

- Where are complaint volumes increasing?
- Which customers demonstrate repeat complaint patterns?
- Which customers have unsatisfied complaints?
- Which customers show service deterioration?

9.5 Executive Management

- What is the current health of the bank?
- What are the most significant business risks?
- Which customers require immediate attention?
- What actions should business teams take?
- Which stakeholders need to be notified?

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10. Dashboard Blueprint

Dashboard	Purpose	Primary Users
Executive Banking Overview	Provide an executive view of banking performance, customer health, financial exposure and key risk indicators.	CEO, Executive Management
Customer 360 / Customer Intelligence	Provide a unified customer view across profile, accounts, financial activity, credit behaviour and service experience.	Retail Banking, Relationship Managers
Loan Portfolio & Credit Risk	Monitor lending exposure, repayment behaviour, DPD, missed payments and credit deterioration.	Head of Credit Risk, Credit Teams
Branch Performance	Monitor branch-level performance, customer activity and operational indicators.	Regional Managers, Branch Managers
Complaint Intelligence	Analyze complaint volume, satisfaction, repeat complaints and service deterioration.	Customer Service, Branch Management
Early Warning Intelligence	Identify, explain and prioritize customers requiring intervention.	Executive Management, Credit Risk, Retail Banking, Operations

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11. Early Warning Intelligence Requirements

The Early Warning layer shall translate customer-level signals into explainable warning classifications.

Element	Requirement
Customer Health	Classify customers into business-health categories such as Healthy, Watchlist and At Risk.
Warning Reason	Identify the primary business reason associated with the warning.
Warning Severity	Classify warning severity, including High and Medium conditions where applicable.
Warning Type	Classify the warning as Credit Deterioration, Service Deterioration or Credit + Service.
Recommended Action	Specify the recommended intervention, such as Relationship Manager Follow-up, Customer Service Follow-up or Service Recovery.
Stakeholder	Identify the business team responsible for intervention.
Stakeholder Email	Provide the notification destination for the responsible stakeholder.
Financial Exposure	Provide outstanding principal associated with the alert.
Priority	Prioritize cases using warning severity, DPD, complaint volume and outstanding exposure.

12. Stakeholder Routing & Automation Requirements

The platform shall support an operational path from analytical detection to stakeholder notification.

The completed demonstration uses n8n as the workflow automation platform. n8n was selected for the demonstration because it can provide flexible workflow orchestration in a personal development environment without requiring dependence on an organizational Microsoft 365 tenant. Power Automate remains a valid production alternative, particularly where an organization is already standardized on the Microsoft ecosystem.

Automation flow:

Power BI / Early Warning Automation Dataset → CSV Processing → Stakeholder Grouping → Stakeholder-Specific HTML Email → Gmail Notification

The automation layer shall:

- Group alerts by stakeholder and stakeholder email.
- Generate stakeholder-specific alert summaries.
- Include priority customers and key risk indicators in the notification.
- Deliver notifications through email.
- Support a clear path from detection to recommended action.

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13. Success Criteria

The project is considered successful when:

- Executives can monitor overall banking health and key risks from centralized Power BI reporting.
- Business users can investigate customers through Customer 360 intelligence.
- Credit teams can identify repayment and credit deterioration indicators.
- Branch and regional users can evaluate operational performance.
- Customer service teams can identify complaint and service deterioration.
- Users can understand why a customer has been flagged.
- Users receive recommended actions associated with identified warnings.
- Warnings can be routed to the responsible stakeholder.
- The automation demonstration can generate and deliver stakeholder-specific email alerts.
- The solution demonstrates an end-to-end flow from enterprise data to business insight and operational action.

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14. Assumptions & Constraints

14.1 Assumptions

- Every customer has a unique Customer ID.
- Every customer belongs to one primary home branch.
- A customer may own one or more bank accounts.
- Every account belongs to one customer.
- Every loan belongs to one customer.
- Every loan repayment record belongs to one loan.
- Every transaction is associated with a valid customer account.

- Branches operate independently but report to regional management.
- All generated data is synthetic and created solely for demonstration and portfolio purposes.
- Historical data spans January 2021 through December 2025.
- Business events and trends are intentionally simulated to reflect realistic banking operations and support meaningful analytics.
- Early Warning classifications are based on predefined business rules.

14.2 Constraints

- The project uses synthetic data only and contains no real customer or banking information.
- Power BI Desktop is used as the primary reporting and visualization platform.
- Python is used to generate realistic banking datasets.
- SQL Server is used as the enterprise data warehouse and analytical data foundation.
- The solution is developed as an offline analytical platform with no live banking-core integration.
- The full enterprise-scale dataset is used for the completed demonstration, subject to local processing resources.
- Risk scores and Early Warning indicators are based on predefined business rules rather than machine-learning models.
- The automation layer is a demonstration implementation rather than a production banking notification service.

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15. Technology Requirements

Layer	Technology
Data Generation	Python, Pandas, Faker
Data Warehouse	Microsoft SQL Server
ETL / Data Transformation	T-SQL
Analytics / Semantic Layer	SQL Views, Stored Procedures, Power BI, DAX
Business Intelligence	Microsoft Power BI
Workflow Automation	n8n
File Storage / Workflow Input	Google Drive
Email Notification	Gmail
Version Control	Git / GitHub

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16. Data Volume & Scale

Component	Approximate Volume
Customers	100,000
Accounts	153,035
Branches	150
Calendar Dates	3,834
Transactions	100M+
Active Loans	4,937
Repayment Records	33,000+
Complaint Records	1.13M+
Early Warning Automation Records	47,847
Stakeholder Routing Groups	13

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17. Final Business Outcome

EBIEWP delivers a completed Business Intelligence and decision-intelligence demonstration that connects enterprise data engineering with business action.

The platform does not stop at reporting. It provides a structured mechanism for identifying customers requiring attention, explaining the warning condition, prioritizing the case, recommending an intervention, identifying the responsible stakeholder and demonstrating automated notification.

The resulting business process is:

Detect → Understand → Prioritize → Recommend → Route → Notify → Act

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18. Document Status

This document represents the final Business Requirements Document for the completed EBIEWP portfolio implementation.

Technical implementation details, dashboard evidence, automation implementation and project architecture are maintained separately in the project's consolidated technical documentation.